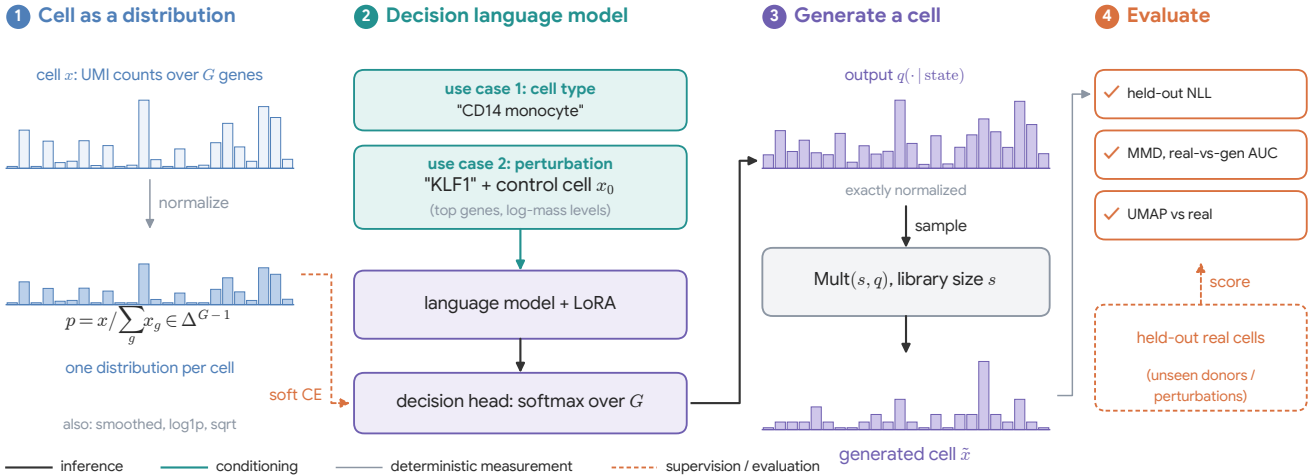
1 Cell as a distribution
cell x: UMI counts over G genes
normalize
$p = x / \sum_g x_g \in \Delta^{G-1}$
one distribution per cell
also: smoothed, log1p, sqrt
2 Decision language model
use case 1: cell type
"CD14 monocyte"
use case 2: perturbation
"KLF1" + control cell x_0
(top genes, log-mass levels)
language model + LoRA
decision head: softmax over G
soft CE
3 Generate a cell
output $q(\cdot \mid \text{state})$
exactly normalized
sample
Mult(s, q), library size s
generated cell $\tilde{x}$
4 Evaluate
held-out NLL
MMD, real-vs-gen AUC
UMAP vs real
score
held-out real cells
(unseen donors / perturbations)
inference
conditioning
deterministic measurement
supervision / evaluation